

Exact Traces and Escape Ranks for a Three-Symbol Open Walsh Gate

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Abstract

We study a finite three-symbol Walsh gate obtained by passing one qutrit per step through a rank-two Fourier–projection contraction. The gate has exact left and right defect projections, a tensor-power identity after one full factor cycle, and an all-period trace formula governed by a greatest common divisor. Newton recursion then gives exact secular polynomials. A tempting one-step rank formula is false: the rank is $2 \cdot 3^{k-1}$, while 2^k occurs only after k steps. The construction is a finite scattering subgate with a closed unitary parent, not a self-adjoint or target-matched quantization.

1 Frozen open gate

Let $\omega = e^{2\pi i/3} = (-1 + i\sqrt{3})/2$ and

$$(F_3)_{j\ell} = 3^{-1/2} \omega^{j\ell}, \quad P = \text{diag}(1, 0, 1), \quad A = F_3^* P.$$

On $\mathcal{H}_k = (\mathbb{C}^3)^{\otimes k}$, with lexicographic qutrit basis, freeze

$$B_k(v_0 \otimes \cdots \otimes v_{k-1}) = v_1 \otimes \cdots \otimes v_{k-1} \otimes Av_0. \quad (1)$$

One application is one clock tick, and $D_k(z) = \det(I_{3^k} - zB_k)$. No spectral rescaling is performed.

Theorem 1. *For every $k, n \geq 1$, with $d = \gcd(n, k)$,*

$$\begin{aligned} \|B_k\| &= 1, & \text{rank } B_k &= 2 \cdot 3^{k-1}, & B_k^k &= A^{\otimes k}, & \text{rank } B_k^k &= 2^k, & (2) \\ \text{Tr}(B_k^n) &= \text{Tr}(A^{n/d})^d. & & & & & & & (3) \end{aligned}$$

The subunitarity defects are the rank- 3^{k-1} projections

$$I - B_k^* B_k = (I - P) \otimes I_{3^{k-1}}, \quad (4)$$

$$I - B_k B_k^* = I_{3^{k-1}} \otimes F_3^* (I - P) F_3. \quad (5)$$

Proof. Write $S(v_0, \dots, v_{k-1}) = (v_1, \dots, v_{k-1}, v_0)$ and $C = I^{\otimes(k-1)} \otimes A$. Then $B_k = CS$, while $A^* A = P$ and $AA^* = F_3^* P F_3$. This proves the norm, one-step rank, and both Gram formulas. Tracking a pure tensor for k steps sends every factor through A once and restores its position, proving the tensor identity and its rank.

For (3), multiplication of basis matrix elements identifies factor positions along the cycles of addition by n modulo k . This permutation has d cycles. Each cycle contracts exactly n/d ordered copies of A , and its free qutrit-index sum is $\text{Tr}(A^{n/d})$. The d cycles are independent, giving the displayed power. This proves every period; no finite replay is used. $\square$

The distinction in (2) is substantive. One tick opens only the factor crossing the cut; the full tensor opening occurs only after k ticks.

2 Secular and path formulas

The one-qutrit characteristic polynomial is

$$\det(\lambda I - A) = \lambda(\lambda^2 - \tau\lambda + q), \quad \tau = \frac{\sqrt{3}}{6} - \frac{i}{2}, \quad q = -\frac{1}{2} - \frac{\sqrt{3}i}{6}.$$

Put $t_0 = 2, t_1 = \tau$ and $t_m = \tau t_{m-1} - q t_{m-2}$. Thus $t_m = \text{Tr}(A^m)$. If $D_k(z) = \sum_j c_{k,j} z^j$, then

$$c_{k,0} = 1, \quad c_{k,m} = -\frac{1}{m} \sum_{j=1}^m c_{k,m-j} t_{j/\gcd(j,k)}^{\gcd(j,k)}. \quad (6)$$

This is a reproducible expression in $\mathbb{Q}(\sqrt{3}, i)$. In particular,

$$\begin{aligned} D_1(z) &= 1 + \left(-\frac{\sqrt{3}}{6} + \frac{i}{2}\right) z + \left(-\frac{1}{2} - \frac{\sqrt{3}i}{6}\right) z^2, \\ D_2(z) &= 1 + \left(-\frac{\sqrt{3}}{6} + \frac{i}{2}\right) z + \left(-\frac{\sqrt{3}}{6} + \frac{i}{6}\right) z^3 + \left(-\frac{1}{6} - \frac{\sqrt{3}i}{6}\right) z^4. \end{aligned}$$

The exact payload lists every coefficient through $k = 5$.

In the frozen basis, a directed edge carries its complex matrix entry. Matrix multiplication and primitive-root decomposition give

$$\text{Tr}(B_k^n) = \sum_{w \text{ rooted closed}} a(w), \quad D_k(z) = \prod_{[p] \text{ primitive}} (1 - a(p) z^{|p|}). \quad (7)$$

All signed and complex cancellations are retained. Since the maximum absolute column sum is $\sqrt{3}$, the raw product is absolutely regroupable for $|z| < 1/\sqrt{3}$. The determinant is globally a polynomial; no larger raw product domain is claimed.

3 Controls and boundary

For $P = I_3$, $A = F_3^*$ and the corresponding B_k is unitary of rank 3^k , with vanishing defects. This is the closed parent from which the opening is defined. Moving the hole to $P_0 = \text{diag}(0, 1, 1)$ preserves the opening rank but changes $\text{Tr}(A)$, hence changes every secular polynomial already in its linear coefficient.

Exact replay checks 60 traces and all coefficients for $k = 1, \dots, 5$. The $k = 2$ path prefix through period eight contains 510 rooted closed paths and 71 primitive cycles when period totals are summed. The prefix is not the proof of (2), (3), or (7).

The strict Route-A tuple is (A1_WEAK, A2_FAIL, A3_FAIL, A4_UNITARY_OR_SCATTERING_CANDIDATE), overall ROUTE_A_EXPLORATION. This finite- k subunitary gate is not self-adjoint and has no semiclassical target matching. We claim no target divisor or counting law, prime-like map, arithmetic local data, Euler factor, root number, automorphy, antiunitary symmetry, Hilbert–Pólya operator, or Route-B authorization. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.